Lecture 14: Introduction to Interior Point Method

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Announcement

- Homework 4 due today (3/15) at noon
- Homework 5 posted, due next Wednesday (3/22) at noon

Midterm Exam

- Mar 25th (Saturday) 10:00am-11:30am. Location and Seating will be announced later.
- One piece of A4 note is allowed. Calculator or any other electronic device is not allowed.
- Sample midterm (and its solution) will be posted. Review slides will be posted
- For those who cannot make the midterm, please send an email by Mar 18th evening to me, even if you have already sent me email earlier. The email should contain the reason (with proof) for not being able to attend the exam, the unavailable period and other explanations.

Recap: Sensitivity Analysis

Let

$$\begin{aligned} V = \min_{\mathbf{x}} \quad & \mathbf{c}^T\mathbf{x} \\ \text{s.t.} \quad & A\mathbf{x} = \mathbf{b} \\ & \mathbf{x} \geq 0 \end{aligned}$$

Then $\nabla V(\mathbf{b}) = \mathbf{y}^*$ if the dual problem has a unique optimal solution $\mathbf{y}^*$ and $\nabla V(\mathbf{c}) = \mathbf{x}^*$ if the primal problem has a unique optimal solution $\mathbf{x}^*$.

- If we change b_i by a small amount Δb_i, then the change of objective value will be $\Delta b_i y_i^*$
- If we change c_i by a small amount Δc_i, then the change of objective value will be $\Delta c_i x_i^*$
- The results hold for general LPs (not only standard form)

Sensitivity Analysis

The above analysis only holds for small changes

- Small change means that the optimal basis does not change
- If the optimal basis changes, then the change of optimal value and solution may be quite different, usually requires much more efforts

It is useful to know what range of change is a "small change" so that the local sensitivity analysis will work

Sensitivity Analysis

Case 1: $\mathbf{b}$ changes to $\tilde{\mathbf{b}} = \mathbf{b} + \Delta\mathbf{b}$

- The reduced costs do not change
- Therefore, in order to make sure that the optimal basis is unchanged, we only need to guarantee that

$$\tilde{\mathbf{x}}_B = A_B^{-1}(\mathbf{b} + \Delta\mathbf{b}) = \mathbf{x}^* + A_B^{-1}\Delta\mathbf{b} \geq 0$$

- If we only change one component of $\mathbf{b}$ by λ, then we can solve the range of λ from a set of linear inequalities

Sensitivity Analysis Continued

Case 2: $\mathbf{c}$ changes to $\tilde{\mathbf{c}} = \mathbf{c} + \Delta\mathbf{c}$

- The basic solution (corresponding to the same basis) does not change
- Therefore, in order to make sure that the current basis is still optimal, we only need to guarantee that the reduced costs are still non-negative, i.e.,

$$\tilde{\mathbf{c}}_N - A_N^T (A_B^{-1})^T \tilde{\mathbf{c}}_B \geq 0$$

  We only need to compute the reduced costs for the non-basic part, since the reduced costs for the basic part will still be 0
- If we only change one component of $\mathbf{c}$ by λ, then we can solve the range of λ from a set of linear inequalities

What if the Change is Outside the Range

If the change of $\mathbf{c}$ is so much that the reduced cost of the current solution contains negative number, then

- We can continue with the simplex tableau until it reaches optimal solution.

If the change of $\mathbf{b}$ is so much that the solution corresponding to the original optimal basis B is no longer feasible

- Then we may need to solve the problem from the start
- It can be viewed as that the objective coefficients of the dual problem changed. Then one can use the method that deals with changes in objective coefficients

Changes to A

If the change is for a number in a non-basic column, say A_j, then the original optimal solution is still feasible, the only change is to the reduced cost of jth variable.

- Recompute $\bar{c}_j$. If it is still non-negative, then the original optimal solution stays optimal. Otherwise, update the tableau for the jth column as well as the reduced cost and continue from there.

If the change is for a number in a basic column, then nearly all the numbers in the tableau will change. In general, there is not a simple way to deal with it.

Other Changes

Adding a variable (the rest are kept the same):

- The original BFS is still BFS, the reduced cost is unchanged.
- Only need to check the reduced cost corresponding to the new variable. If it is non-negative, then the original optimal solution is still optimal; otherwise continue the simplex method from there.

Adding a constraint:

- If the original optimal solution satisfies the constraint, then it is still optimal.
- If not, then the best way to deal with it is to think it as adding a dual variable, then use the simplex tableau for the dual problem to continue calculations.

Start: A New Perspective of Solving LPs

Now we have studied some duality theories of linear optimization. Let's review the simplex method from a new perspective.

We start with the optimality conditions of LP (consider the standard form):

1. (Primal Feasible) $A\mathbf{x} = \mathbf{b}$, $\mathbf{x} \geq 0$
2. (Dual Feasible) $A^T\mathbf{y} \leq \mathbf{c}$
3. (Complementarity) $x_i \cdot s_i = x_i \cdot (c_i - A_i^T\mathbf{y}) = 0$ for each i

They are necessary and sufficient conditions for the optimal solution of the primal and dual problems.

Review of the Simplex Method

In the simplex method, we search among basic feasible solutions.

- Therefore, we always keep primal feasibility

For any basis B, define $\mathbf{y} = (A_B^{-1})^T \mathbf{c}_B$. Note that $\mathbf{c}^T - \mathbf{y}^T A$ is the reduced cost. And we have for basic variables, the reduced costs are zero. Therefore,

$$x_i \cdot (c_i - A_i^T \mathbf{y}) = 0 \quad \forall i$$

- Therefore, we always keep complementarity conditions

Review of the Simplex Method

During the simplex method, the reduced costs may be negative

- That is, the dual solution $\mathbf{y} = (A_B^{-1})^T \mathbf{c}_B$ may not always satisfy $A^T \mathbf{y} \leq \mathbf{c}$
- It stops whenever the reduced costs are non-negative
- We seek $\mathbf{y}$ that satisfies $A^T \mathbf{y} \leq \mathbf{c}$, i.e., dual feasibility

Proposition

During each iteration, the simplex method maintains primal feasibility and complementarity conditions, and seeks a solution that is dual feasible.

What if we choose two other conditions to maintain?

- It will result in dual-simplex method and interior point method

Dual Simplex Method

One can view the dual simplex method to be a simplex method applied to the dual problem of an LP.

- It maintains dual feasibility
- It maintains complementarity conditions
- However, it doesn't need to be primal feasible during the process. It seeks for primal feasibility

The tableau can be viewed as rotated from the primal one.

There are cases where using a dual simplex method is more convenient, for example

- If there is a dual BFS available (but no primal BFS available)
- We have mentioned one such scenario in the discussion of sensitivity analysis (when $\mathbf{b}$ is changed by a large amount or a constraint is added)

Interior Point Method

Remember the optimality condition

1. (Primal Feasible) $A\mathbf{x} = \mathbf{b}$, $\mathbf{x} \geq 0$
2. (Dual Feasible) $A^T\mathbf{y} \leq \mathbf{c}$
3. (Complementarity) $x_i \cdot s_i = x_i \cdot (c_i - A_i^T\mathbf{y}) = 0$ for each i

Interior point method maintains both primal feasibility and dual feasibility during its iterations and seeks for a pair of solutions that satisfy the complementarity conditions.

High-Level Idea

We want to find $\mathbf{x}$, $\mathbf{y}$ and $\mathbf{s}$ such that

$$\begin{aligned} A\mathbf{x} &= \mathbf{b}, \quad \mathbf{x} \geq 0 \\ A^T\mathbf{y} + \mathbf{s} &= \mathbf{c}, \quad \mathbf{s} \geq 0 \\ x_i \cdot s_i &= 0, \quad \forall i \end{aligned}$$

This is a set of nonlinear equations. It is not very easy to find a solution.

High-Level Idea Continued

We consider a relaxed version of the problem.

$$\begin{aligned} A\mathbf{x} &= \mathbf{b}, \quad \mathbf{x} \geq 0 \\ A^T\mathbf{y} + \mathbf{s} &= \mathbf{c}, \quad \mathbf{s} \geq 0 \\ x_i \cdot s_i &\leq \mu, \quad \forall i \end{aligned}$$

We call $\mu > 0$ the complementarity gap.

Idea: If we have found a solution for a certain μ, then we might be able to find a solution for a smaller μ. Then we keep decreasing μ and finally we can get the solution to the LP

- The essential steps in the interior point method is to show that this is possible to do — the method is similar to the Newton's method, which we will discuss in the second half of the semester.

Why Called Interior Point Method?

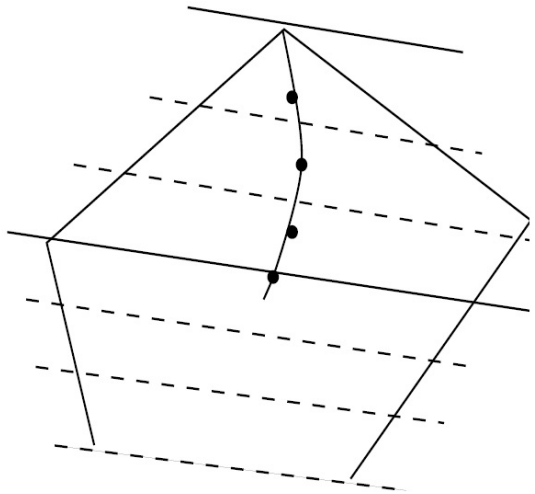


Figure: Central path in an interior point method

- In the simplex method, we only search among the extreme points (all on the boundaries of the polytope)
- In the interior point method, we search in the interior of the feasible region
- The optimal solution output by interior point method may not be a BFS (if the solution is unique, then it must be a BFS)

Interior Point Method

Start with any feasible solution $\mathbf{x}$, $\mathbf{y}$ and $\mathbf{s}$ satisfying

$$\begin{aligned} A\mathbf{x} &= \mathbf{b}, \quad \mathbf{x} > 0 \\ A^T\mathbf{y} + \mathbf{s} &= \mathbf{c}, \quad \mathbf{s} > 0 \\ x_i \cdot s_i &= \mu_i, \quad \forall i \end{aligned}$$

Now we want to find the next solution $\bar{\mathbf{x}} = \mathbf{x} + \mathbf{dx}$, $\bar{\mathbf{y}} = \mathbf{y} + \mathbf{dy}$ and $\bar{\mathbf{s}} = \mathbf{s} + \mathbf{ds}$ such that:

$$\begin{aligned} A(\mathbf{x} + \mathbf{dx}) &= \mathbf{b}, \quad \bar{\mathbf{x}} > 0 \\ A^T(\mathbf{y} + \mathbf{dy}) + (\mathbf{s} + \mathbf{ds}) &= \mathbf{c}, \quad \bar{\mathbf{s}} > 0 \\ (x_i + dx_i) \cdot (s_i + ds_i) &= \eta\mu_i, \quad \forall i \end{aligned}$$

where η is a constant less than 1.

Interior Point Method

$$\begin{aligned} A(\mathbf{x}+\mathbf{dx}) &= \mathbf{b}, \quad \bar{\mathbf{x}} > 0 \\ A^T(\mathbf{y}+\mathbf{dy}) + (\mathbf{s}+\mathbf{ds}) &= \mathbf{c}, \quad \bar{\mathbf{s}} > 0 \\ (x_i + dx_i)\cdot(s_i + ds_i) &= \eta\mu_i, \quad \forall i \end{aligned}$$

This is a system of nonlinear equations. If we can solve it, we can continue the procedure until it reaches a solution with $\mu_i = 0$.

Important observation

- Everything is linear except the $dx_i \cdot ds_i$ term in the last equation
- If we keep our step size small, then that term will be negligible (second-order quantity)

Interior Point Method

We solve dx_i, ds_i that satisfies:

$$\begin{aligned} A\mathbf{dx} &= 0 \\ A^T\mathbf{dy} + \mathbf{ds} &= 0 \\ s_i dx_i + x_i ds_i &= \eta\mu_i - \mu_i \quad \forall i \end{aligned}$$

There are $2n + m$ unknowns, and $2n + m$ equations, so there exists a unique solution

Interior Point Method

We then choose the next solution to be

$$\mathbf{x}^+ = \mathbf{x} + \alpha \mathbf{dx}$$
$$\mathbf{y}^+ = \mathbf{y} + \alpha \mathbf{dy}$$
$$\mathbf{s}^+ = \mathbf{s} + \alpha \mathbf{ds}$$

By choosing α sufficiently small, we can guarantee that

- $\mathbf{x}^+$ is primal feasible
- $(\mathbf{y}^+, \mathbf{s}^+)$ is dual feasible
- Each $x_i^+ \cdot s_i^+$ will be smaller than μ_i

Then we keep shrinking μ_i, we will get a sequence of solutions with smaller and smaller complementarity gaps

Interior Point Method

We can also prove that when μ is small enough, if we can get a solution that satisfies:

$$\begin{aligned} A\mathbf{x} &= \mathbf{b}, \quad \mathbf{x} \geq 0 \\ A^T\mathbf{y} + \mathbf{s} &= \mathbf{c}, \quad \mathbf{s} \geq 0 \\ x_i \cdot s_i &\leq \mu, \quad \forall i, \end{aligned}$$

then we can identify an exact optimal solution.

- We can prove that this algorithm will converge to an optimal solution after $O(\sqrt{n})$ iterations.
- $O(\sqrt{n})$ means a number between $C_1\sqrt{n}$ and $C_2\sqrt{n}$ where C_1 and C_2 are constants that do not depend on n.

Initialization of Interior Point Method

In the simplex method, we need an initial BFS to start.

- In the interior point method, we need to have an initial interior point to start.

We need $\mathbf{x}$, $\mathbf{y}$ and $\mathbf{s}$ that satisfy:

$$\begin{aligned} A\mathbf{x} &= \mathbf{b}, \quad \mathbf{x} > 0 \\ A^T\mathbf{y} + \mathbf{s} &= \mathbf{c}, \quad \mathbf{s} > 0 \end{aligned}$$

Like in the simplex method, it is not trivial.

Initialization of Interior Point Method

The current method is to consider an augmented problem called *homogeneous self-dual problem* (HSDP):

$$
\begin{array}{rlllll}
\text{minimize}_{\mathbf{x},\mathbf{y},\tau,\theta} & & & (n+1)\theta & & \\
\text{s.t.} & A\mathbf{x} & -\mathbf{b}\tau & +\bar{\mathbf{b}}\theta & = & 0 \\
-A^T\mathbf{y} & & +\mathbf{c}\tau & -\bar{\mathbf{c}}\theta & \geq & 0 \\
\mathbf{b}^T\mathbf{y} & -\mathbf{c}^T\mathbf{x} & & +\bar{z}\theta & \geq & 0 \\
-\bar{\mathbf{b}}^T\mathbf{y} & +\bar{\mathbf{c}}^T\mathbf{x} & -\bar{z}\tau & & = & -(n+1) \\
\mathbf{y} \text{ free}, & \mathbf{x} \geq 0, & \tau \geq 0 & \theta \text{ free} & &
\end{array}
$$

where $\bar{\mathbf{b}} = \mathbf{b} - A\mathbf{e}$, $\bar{\mathbf{c}} = \mathbf{c} - \mathbf{e}$, $\bar{z} = \mathbf{c}^T\mathbf{e} + 1$.

- This problem is self-dual, i.e., the dual is equivalent to itself.
- This problem has a natural initial interior point ($\mathbf{y} = 0$, $\mathbf{x} = \mathbf{e}$, $\tau = 1$, $\theta = 1$)
- Solving this problem will give the optimal solution to the original problem (if exist)

Interior Point Method

Complexity: Interior point method is a polynomial-time algorithm with overall complexity roughly $O(n^{3.5})$

- There are several variants of the interior point method. The one we introduced is called the *primal-dual* type of interior point method.
- The main ideas of those variants are similar, i.e., going through the *interior* of the feasible region and seeking for complementarity

Practical Performance

On average, the speed of the simplex method is comparable with the speed of the interior point method despite their difference in theoretical complexity

- For some problems, simplex method can finish very fast (within a few iterations); for other problems, simplex method needs some extra time (worst-case exponential)
- In contrast, the running time of interior point method is quite stable, it doesn't vary much from problem to problem (given fixed size)

Practical Performance

Problem	Gurobi 9.5.0 Spx	Gurobi 9.5.0 IPM	COPT 4.0.1 Spx	COPT 4.0.1 IPM
L1_sixm	3	3	1	2
Linf_520c	39	6	10	4
a2864	95	93	54	55
bdry2	681	79	411	104
cont1	7	8	2	2
cont11	8	8	6	6
datt256	173	12	8	3
dlr1	1002	41	2091	67
ex10	3	2	2	1
fhnw-bin1	132	67	89	34
fome13	4	1	2	1
graph40-40	21	16	11	12
irish-e	25	3	18	6
neos	5	5	13	5
neos3	2	2	1	1
neos3025225	89	20	13	17
neos5052403	30	3	8	4
neos5251015	2	5	3	2
ns1687037	193	6	69	4
ns1688926	4	6	6	4
nug08-3rd	10	14	2	3
ShiftGeoMean	**42.42**	**12.38**	**20.84**	**11.28**
Scaled	**3.76**	**1.10**	**1.85**	**1.00**

One Property of Interior Point Method

Theorem

The interior point method will always find the optimal solution with the maximum possible number of non-zeros.

Consider a simple case:

$$\begin{aligned} \text{minimize}_{\mathbf{x}} \quad & x_1 + x_2 + x_3 \\ \text{s.t.} \quad & x_1 + x_2 + x_3 = 1 \\ & x_1, x_2, x_3 \geq 0 \end{aligned}$$

Simplex method will give a BFS as optimal (one of $(1, 0, 0)$, $(0, 1, 0)$ and $(0, 0, 1)$), while the interior point method will output $(1/3, 1/3, 1/3)$.

Therefore..

In the case of multiple optimal solutions..

- If we want a high-rank solution (with the maximum possible non-zeros), then choose the interior point method
- If we want a low-rank solution (with small number of non-zeros), then choose the simplex method

In particular, the highest-rank solution is unique for linear programs, but the lowest-rank solution may not be unique.

- The optimal solution output by the simplex method may depend on the initial solution (as well as the pivoting rules).

High-Rank v.s. Low-Rank

Either could be desirable in practice.

High-rank:

- In the multi-firm alliance problem, we want a unique allocation
- Location problem (e.g., building the fountain), it will give the solution that lies in the center of all solutions.

Low-rank

- In portfolio problems, we want to minimize the number of stocks chosen (reduce the transaction cost).
- In graph problems, we want to choose fewer nodes/edges.
- In other cases, we prefer integer solutions over fractional solutions (given that their objective values are equal). This usually corresponds to low-rank solutions.

If the optimal solution is unique, then both methods return the same solution.

Software Choices

Both simplex method and interior point method are used by major commercial software

- In MATLAB, we can specify which method we want to use in the function called *linprog* (the interior point method is the default)
- Same in COPT/CPLEX/Gurobi
- The default in CVX is interior point method
- Excel uses simplex method